Exercise 2 - Mhluri Maswanganye

(01) SELECT DISTINCT(department)
FROM students;

departments
IT
HR
Finance

(02) SELECT department,
AVG(age) AS avg-age
FROM students
GROUP BY department;

department	average_age
Finance	23.0
HR	22.0
IT	20.5

(03) SELECT department
COUNT (id) AS student-count
FROM student
GROUP BY department
HAVING COUNT(id) >1;

student_count
5

(04) SELECT *
FROM student
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT

(05) SELECT * →

student_id	name	age	departm
2	Bob	22	HR

FROM students
WHERE (department = 'IT' OR department = 'HR')
AND age > 21;

THE courses TABLE

(06) SELECT department,
SUM(credits) AS total_credits
FROM courses
GROUP BY ~~departments~~
HAVING SUM(credits) > 5;

department	total_credits
IT	11

(07). SELECT course_id,
course_name,
department,
credits
FROM courses
WHERE credits != 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

```
08 SELECT course_id,
          course_name,
          credits
   FROM courses
   ORDER BY credits DESC
   LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

THE ENROLLMENTS TABLE

```
09 SELECT MAX(grade) AS max_grade,
          MIN(grade) AS min_grade,
          AVG(grade) AS avg_grade
   FROM enrollments;
```

max_grade	min_grade	avg_grad
90	78	84.6

```
10. SELECT course_id,
           COUNT(enrollment_id) AS enrollment_count
    FROM enrollments
    GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

THE salaries Table

11.
```
SELECT department,
    SUM(salaries) AS total_salary,
    SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
Finance	70000	6000
HR	109 000	7500
IT	122 000	10500

12.
```
SELECT department,
    AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 59000
```

department	avg_salary
Finance	70000
IT	61000

```
⑬ SELECT employee_id,
          name,
          salary,
          bonus,
          (salary + bonus) AS total_compensation
   FROM salaries
   WHERE (salary + bonus) > 6000;
```

⑭

employee_id	name	salary	bonus	total_compansation
1	Tom	60000	5000	65000
3	Spike	70000	6000	7600
4	Tylce	62000	5500	67500

THE PROJECTS TABLE

```
14. SELECT department,
          SUM(budget) AS total_budget,
          AVG(budget) AS avg_budget
   FROM projects
   GROUP BY department
   HAVING AVG(budget) > 70000;
```

department	total_budget	avg_budget
IT	270000	135 000
Finance	80000	80 000

15. SELECT project_id,
project_name,
department,
budget
FROM projects
WHERE budget BETWEEN 50000 AND 120 000
AND department <> 'Marketing';

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000